

Statistical Contraction Theory for Learning-based Control of Nonlinear Systems—Part 0: Introduction

Hiroyasu Tsukamoto

Nonlinear Autonomous Control, Exploration, Intelligence, and Systems (N-ACXIS) Laboratory
Department of Aerospace Engineering
University of Illinois Urbana-Champaign

Caltech



Jet Propulsion Laboratory
California Institute of Technology



**Massachusetts
Institute of
Technology**



**UNIVERSITY OF
ILLINOIS**
URBANA-CHAMPAIGN

Background



- ▣ **Assistant Professor of Aerospace Engineering**
 - Nonlinear Control Theory (and beyond)



Jet Propulsion Laboratory
California Institute of Technology

- ▣ **Research Affiliate in Robotics**
 - Multi-Agent and System-Level Autonomy



Caltech

- ▣ **Ph.D. (Best Dissertation) & M.S. in Space Engineering**
 - Autonomous Robotics and Control



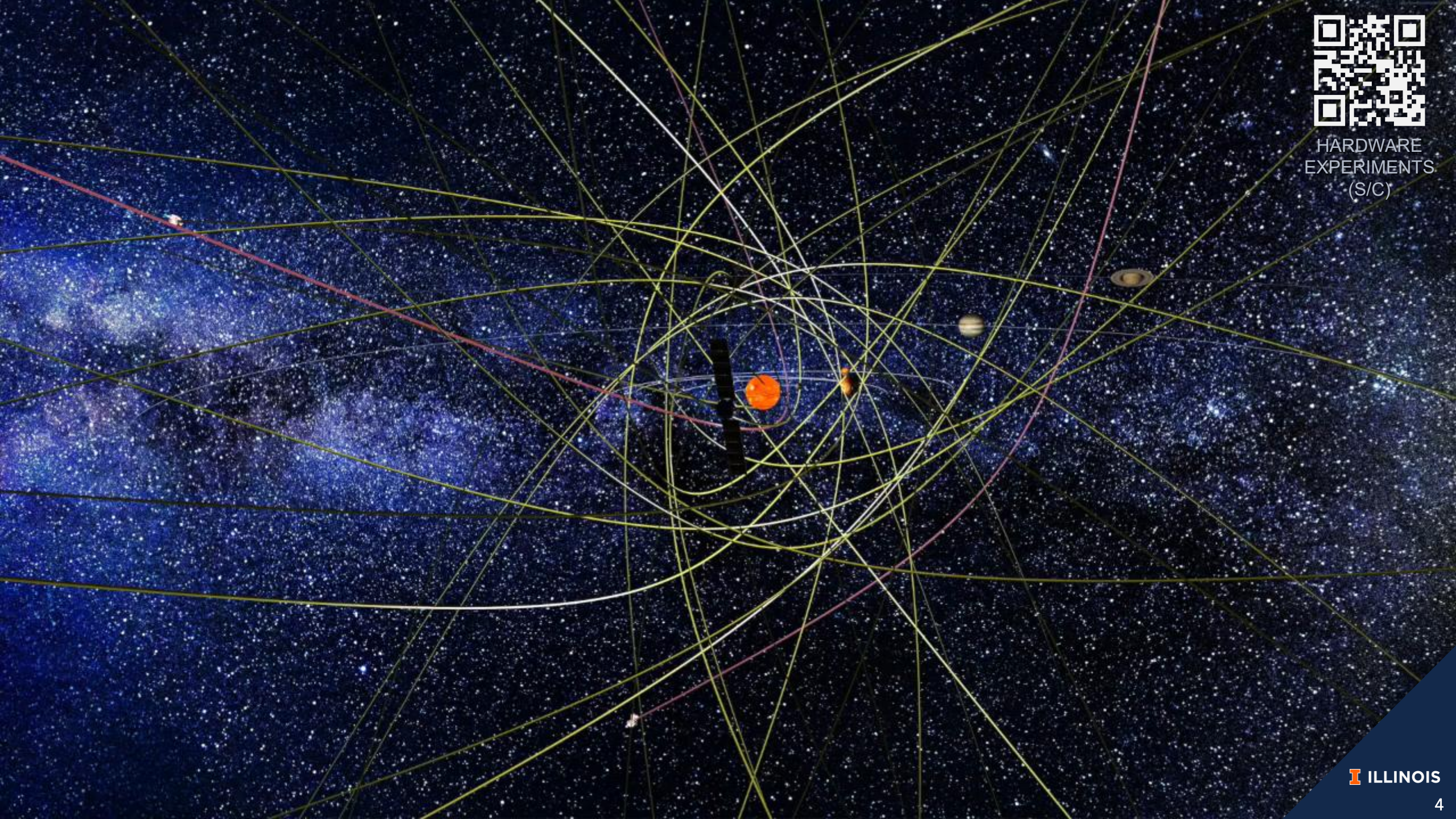
京都大学
KYOTO UNIVERSITY

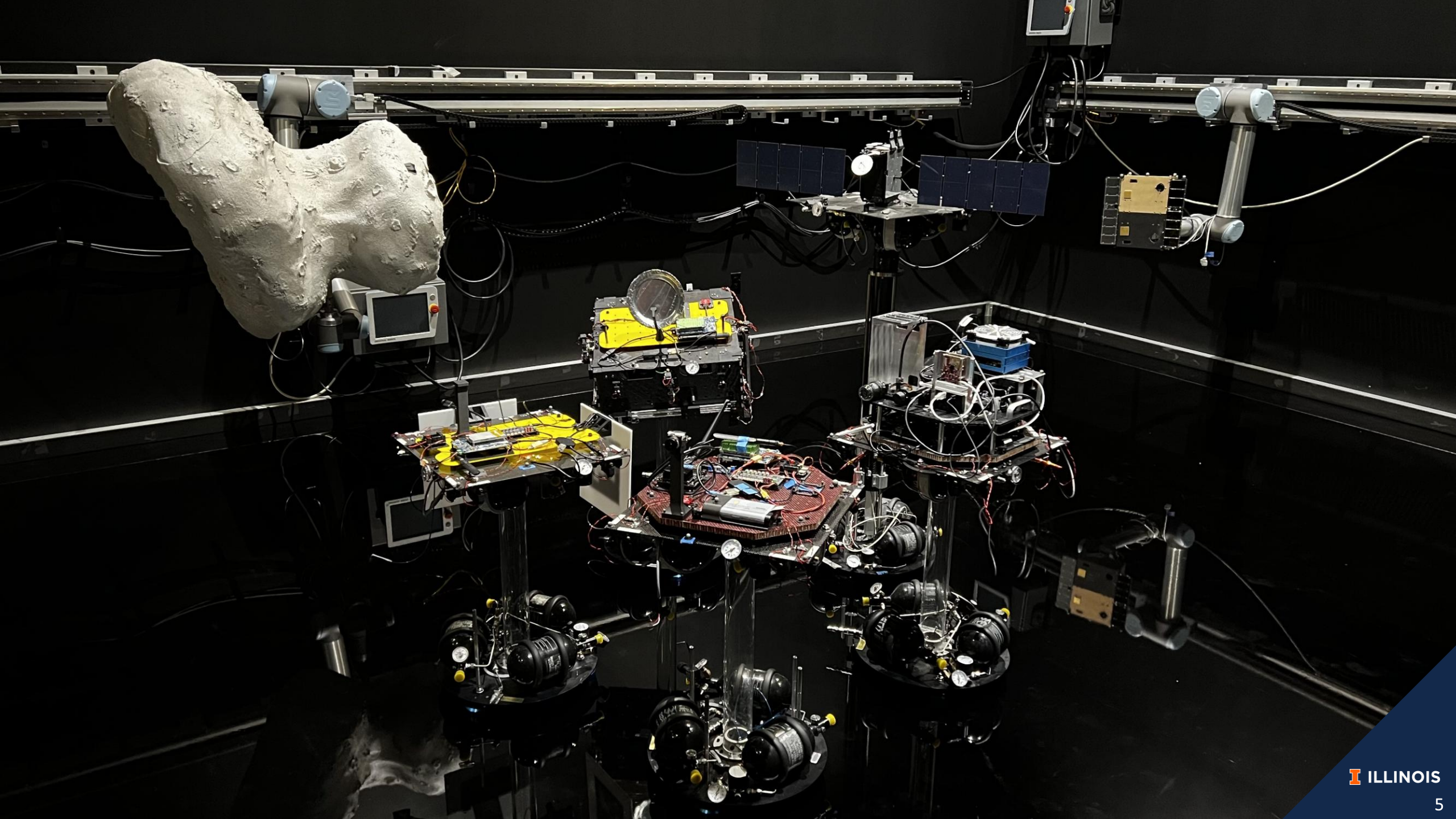
- ▣ **B.S. in Aeronautics and Astronautics**
 - Control and Dynamical Systems

Application Examples



HARDWARE
EXPERIMENTS
(S/C)





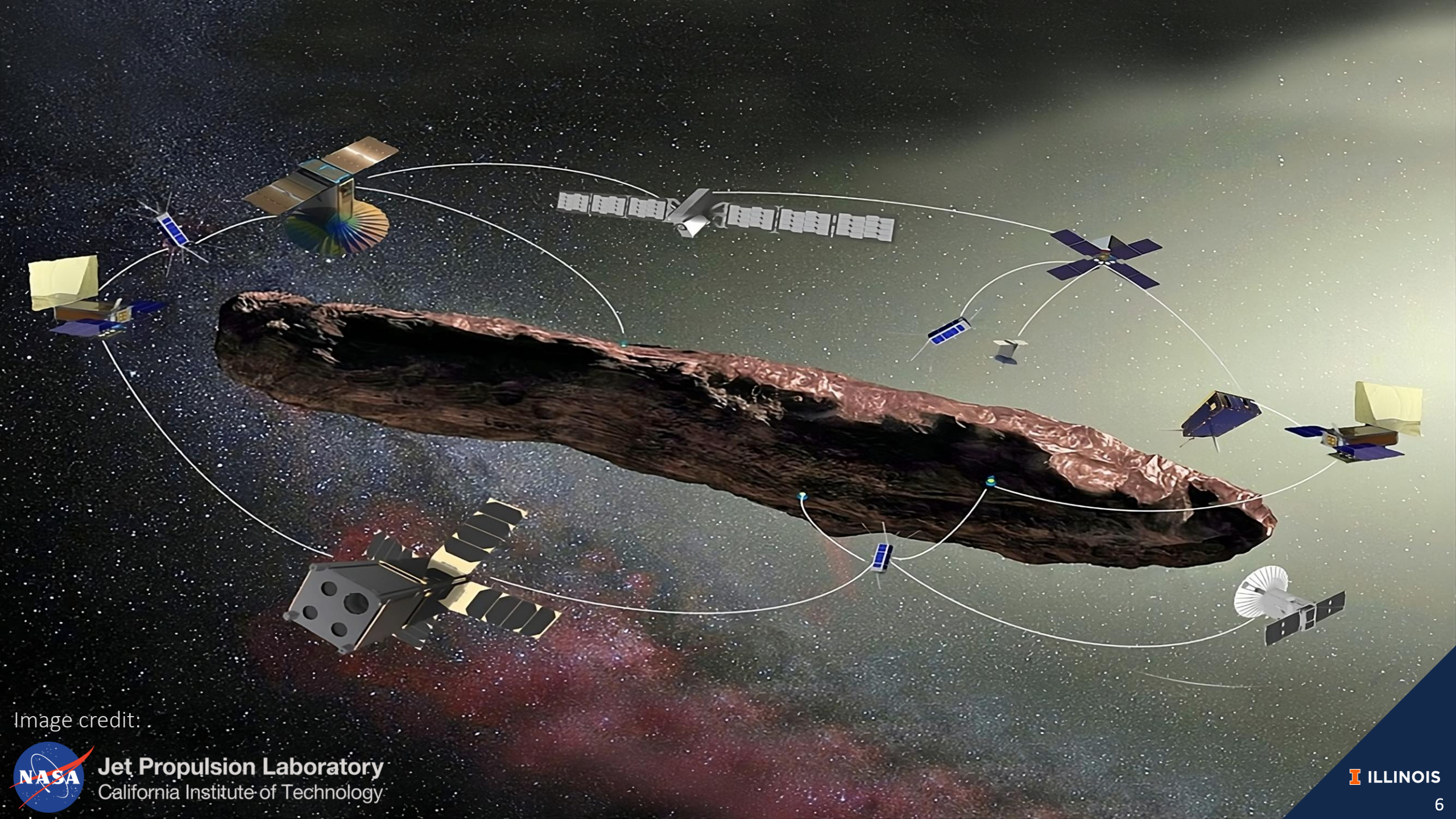


Image credit:

Challenges in Nonlinear Autonomy

Highly nonlinear and large-scale dynamics
Autonomous operations in extreme environments

+

Bounded disturbances
Stochastic uncertainties
Human interactions
Potential faults in systems
Adversarial attacks



Image credit:



120'x60'x20'
mocap space



Onboard & offboard GPU



Bio-inspired flapping UAV



Space-robot prototype (future work)



Micro UAV



Large UAV



FPV UAV

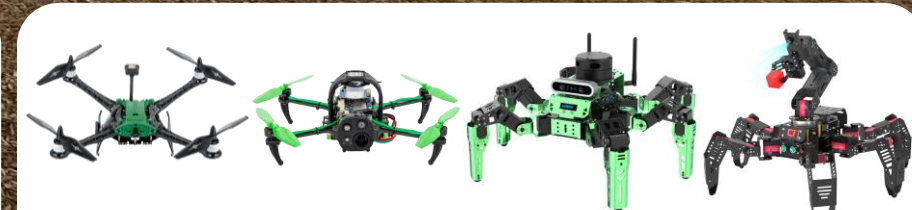
WORLD-LEADING RESEARCH INITIATIVE FOR END-TO-END AUTONOMY IN AEROSPACE ROBOTICS



Nonlinear controlled robot



Ground robot



SLAM and AI-infused UAV and legged robot

Machine learning can hallucinate...



Course Overview

Topics

- Contraction Theory for Nonlinear Stability Analysis and Control
 - Original paper: <https://web.mit.edu/nsl/www/preprints/contraction.pdf>
 - Control contraction metric: <https://arxiv.org/pdf/1503.03144>
 - Recent book: <https://fbullo.github.io/ctds/>
 - Our tutorial paper: <https://arxiv.org/pdf/2110.00675>
- Statistical Inference for Machine Learning
 - PAC Bayes: <https://arxiv.org/pdf/1901.05353>
 - Conformal inference: <https://www.stat.berkeley.edu/~ryantibs/statlearn-s23/lectures/conformal.pdf>
 - Hugging Face: <https://huggingface.co/>
- Data-Driven Contraction Theory
 - Conformal robustness: <https://arxiv.org/pdf/2506.06228>
 - Conformal contraction: <https://arxiv.org/pdf/2507.13613>
 - ⋮

Goals

Simply put, *dynamical systems* = *nominal model* + *uncertainty*

1. Given any *nominal model*, design controllers for nonlinear dynamical systems with contraction guarantees, or understand current theoretical limits
2. Given any *uncertainty*, design learning-based controllers for nonlinear dynamical systems with contraction guarantees, or understand current theoretical limits
3. Identify a *nominal model* and *uncertainty* of your interest, and formulate (and solve) a challenging research problem for your next paper

Potential Research Topics (be ambitious!)

- 1) Foundation Models
- 2) Unconventional Aerospace Robotics
- 3) Symbolic Reasoning and Control
- 4) Generalized Robot Learning
- 5) Safety-Critical Learning and Control
- 6) Robust Learning and Control
- 7) Data-Driven Control
- 8) Intelligent Space Mission Design
- 9) Generalized Alignment in Foundation Models
- 10) Reinforcement Learning
- 11) Your ideas

Potential Research Topics (be ambitious!)

https://canvas.illinois.edu/courses/68555/files/21251458?module_item_id=5078987

Target Conferences and Journals

Conferences

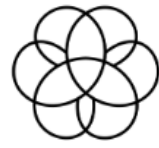
- Conference on Decision and Control (CDC)
- American Control Conference (ACC)
- Learning for Dynamics & Control Conference (L4DC)
- International Conference on Intelligent Robots & Systems (IROS)
- Conference on Robot Learning (CoRL)
- International Conference on Robotics and Automation (ICRA)
- International Conference on Machine Learning (ICML)
- Neural Information Processing Systems (NeurIPS)
- SciTech
- Astrodynamics Specialist Conference
- Space Flight Mechanics Meeting
-

Journals

- IEEE Transactions on Automatic Control
- Automatica
- IEEE Control Systems Letters
- Annual Reviews in Control
- Nature
- Science Robotics
- IEEE Transactions on Robotics
- IEEE Robotics and Automation Letters
- International Journal of Robotics
- AIAA Journal of Guidance, Control, and Dynamics
- Acta Astronautica
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Target Conferences and Journals

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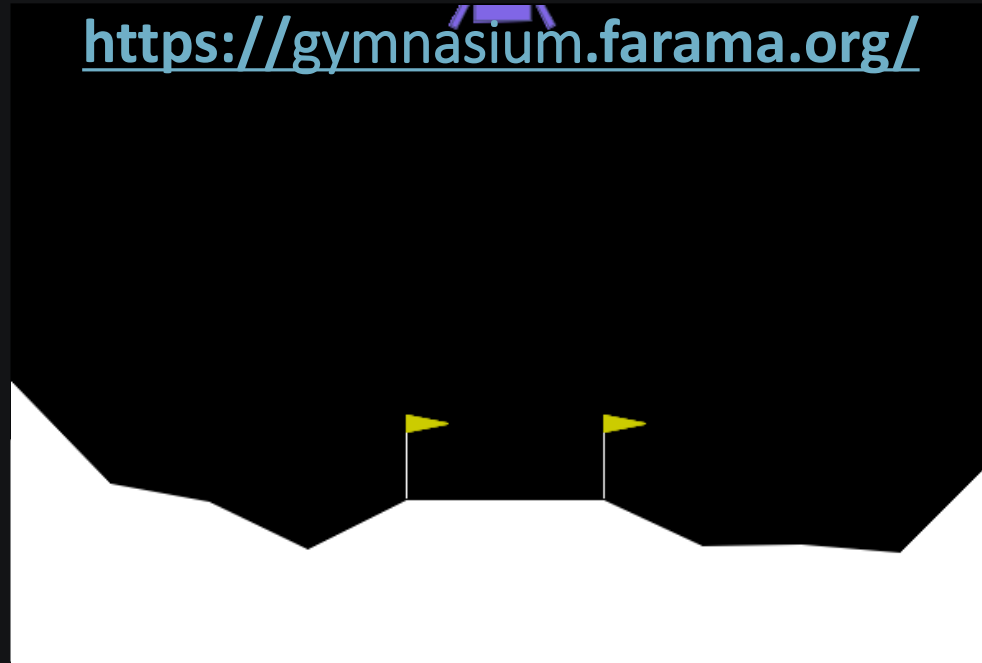


Gymnasium



An API standard for reinforcement learning with a diverse collection of reference environments

<https://gymnasium.farama.org/>



Gymnasium is a maintained fork of OpenAI's Gym library. The Gymnasium interface is simple, pythonic, and capable of representing general RL problems, and has a [compatibility wrapper](#) for old Gym environments:

Transformers and GPU Computations



Transformers and GPU Computations

Python Libraries

- Hugging Face: <https://huggingface.co/docs/hub/en/models-libraries>
- PyTorch: <https://pytorch.org/get-started/locally/>

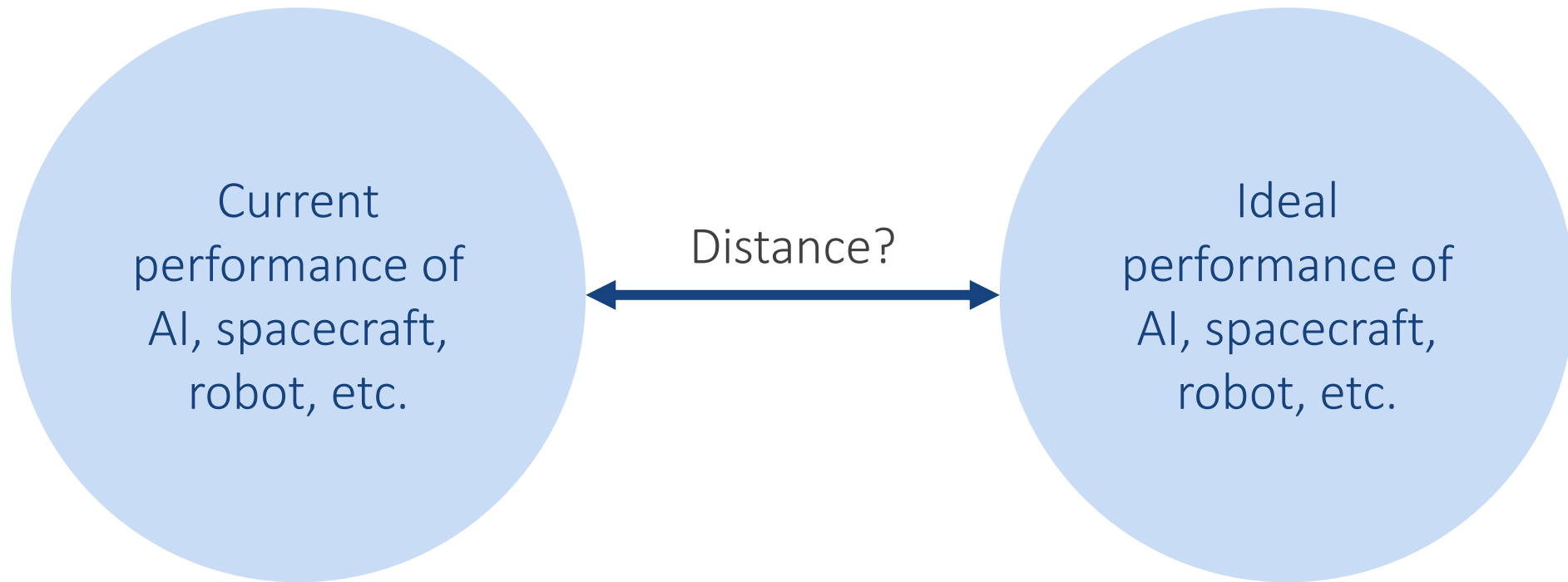
GPU Computations

- PowerEdge
- Delta AI: <https://delta.ncsa.illinois.edu/delta-delta-ai/>

A Brief Look at Technical Details

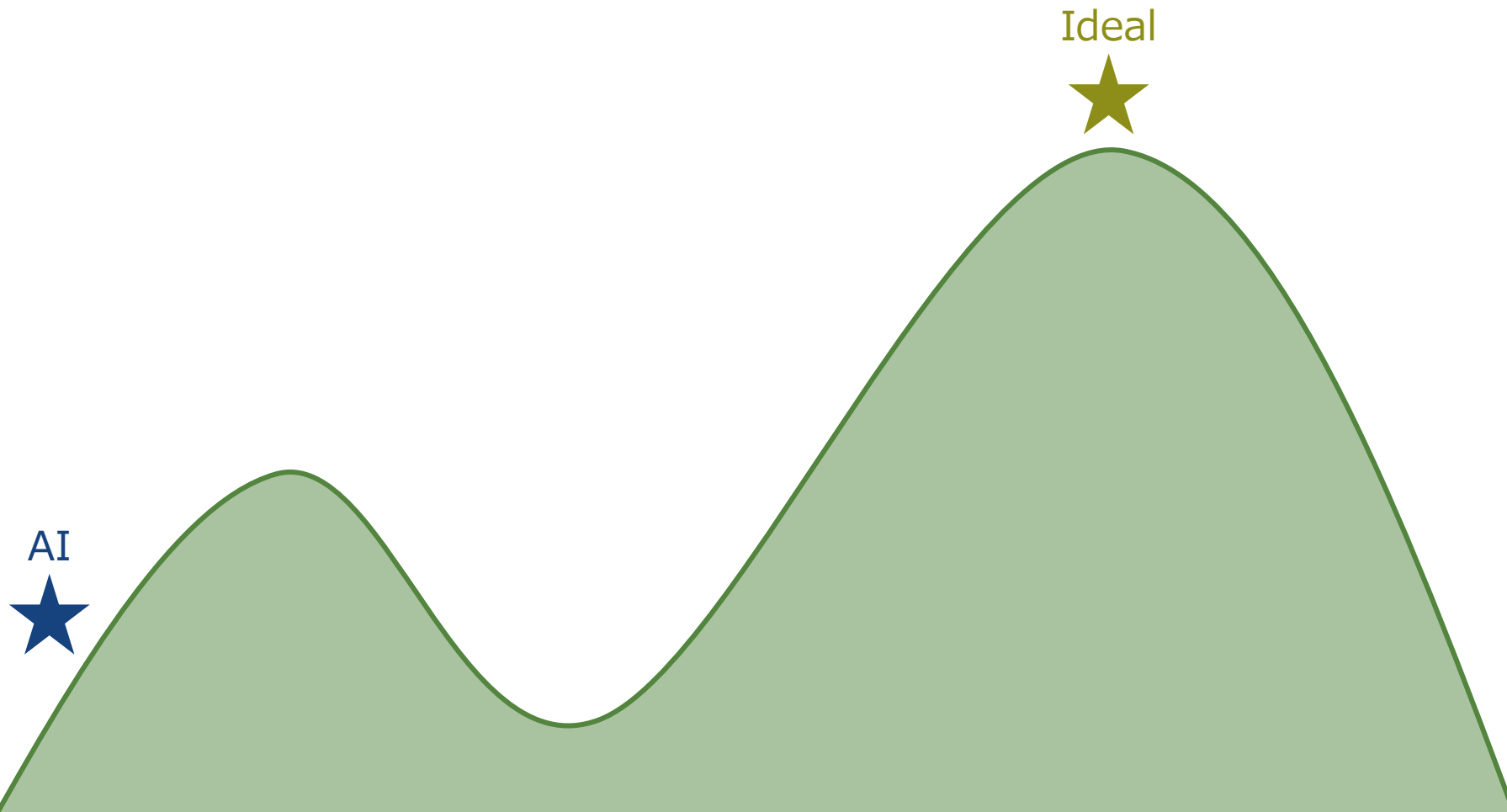
How can we design autonomous systems with guarantees?

Goal: Distance $\rightarrow 0$



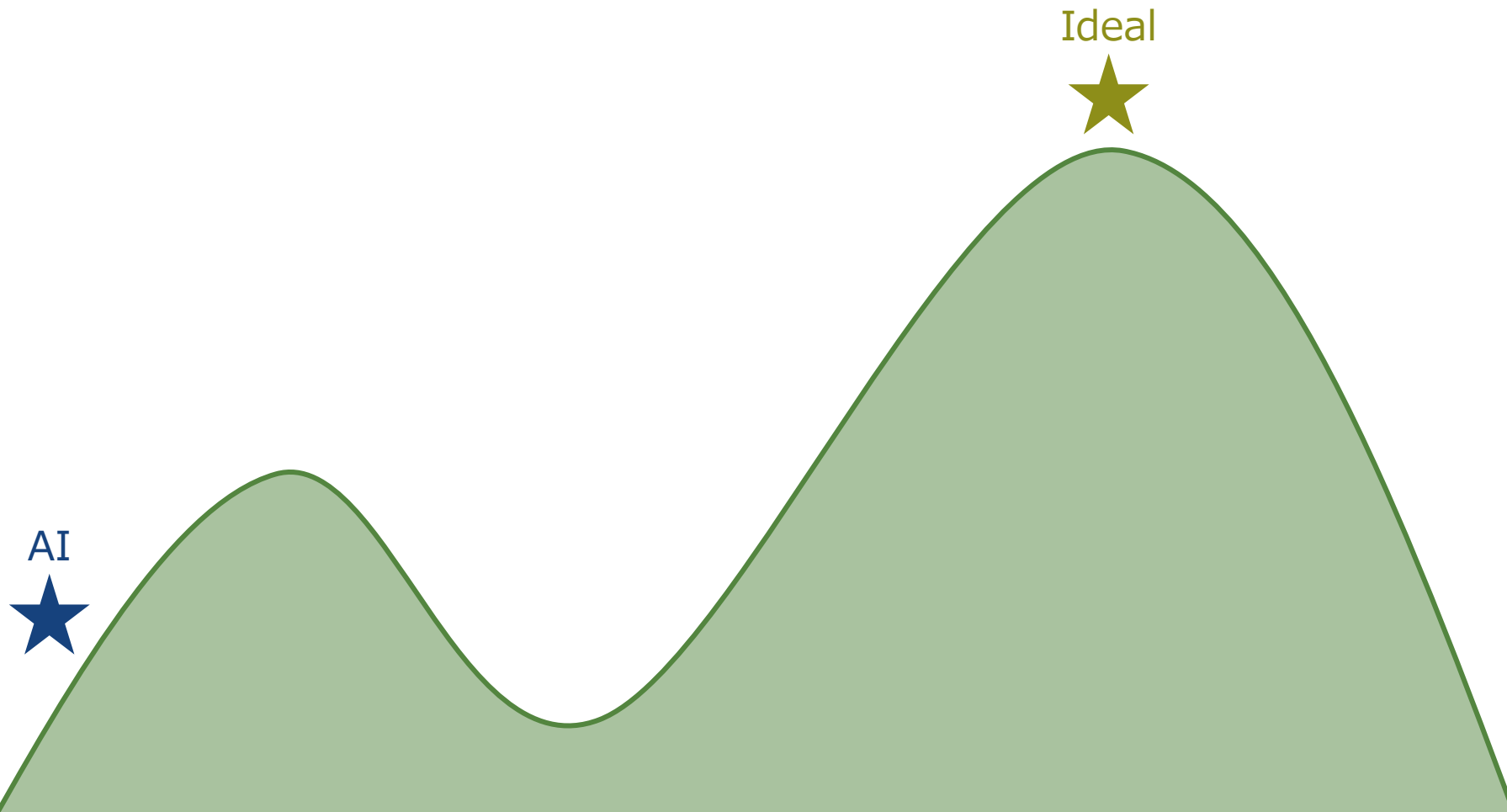
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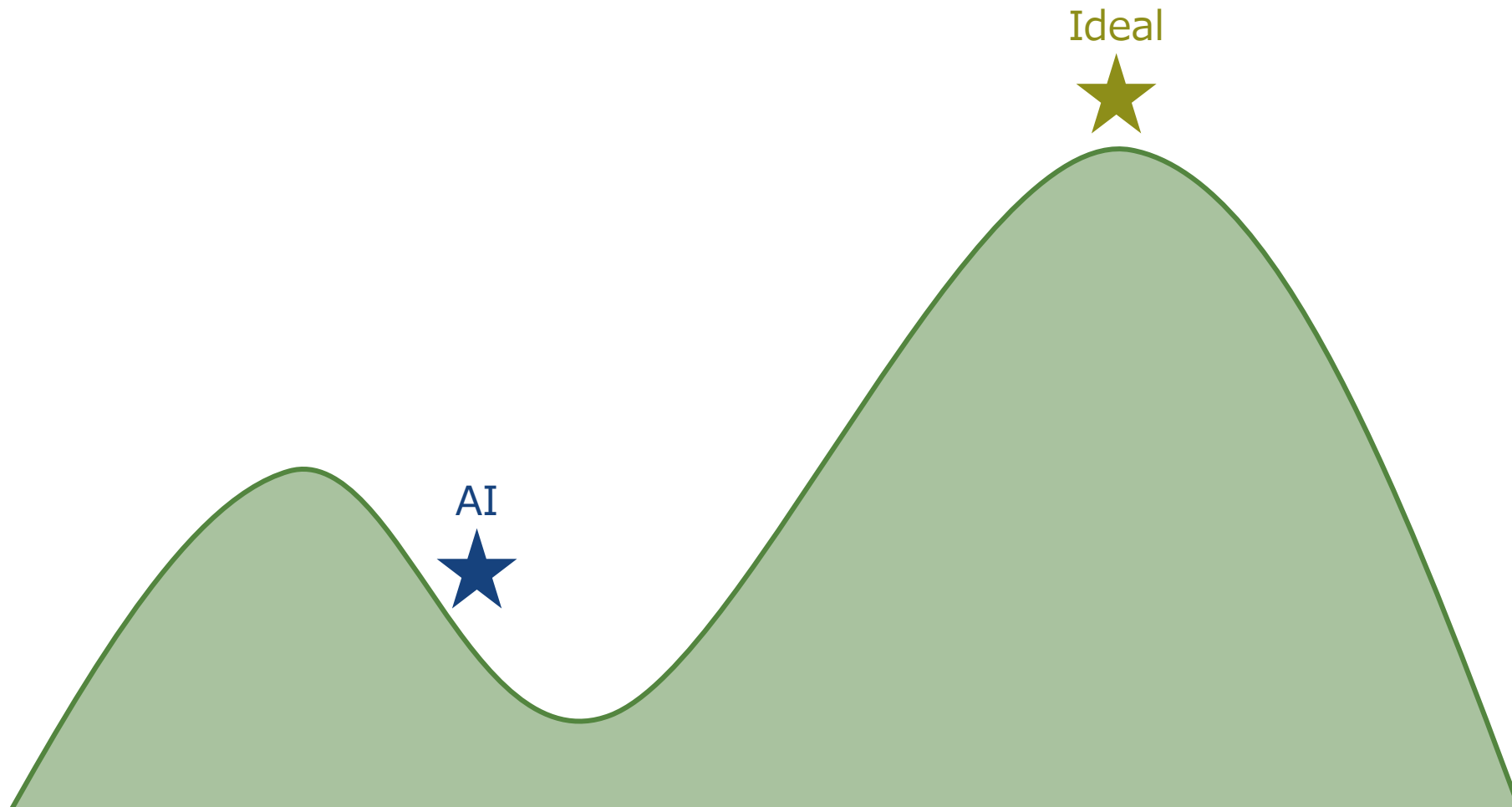
How can we design autonomous systems with guarantees?

If altitude = distance ...



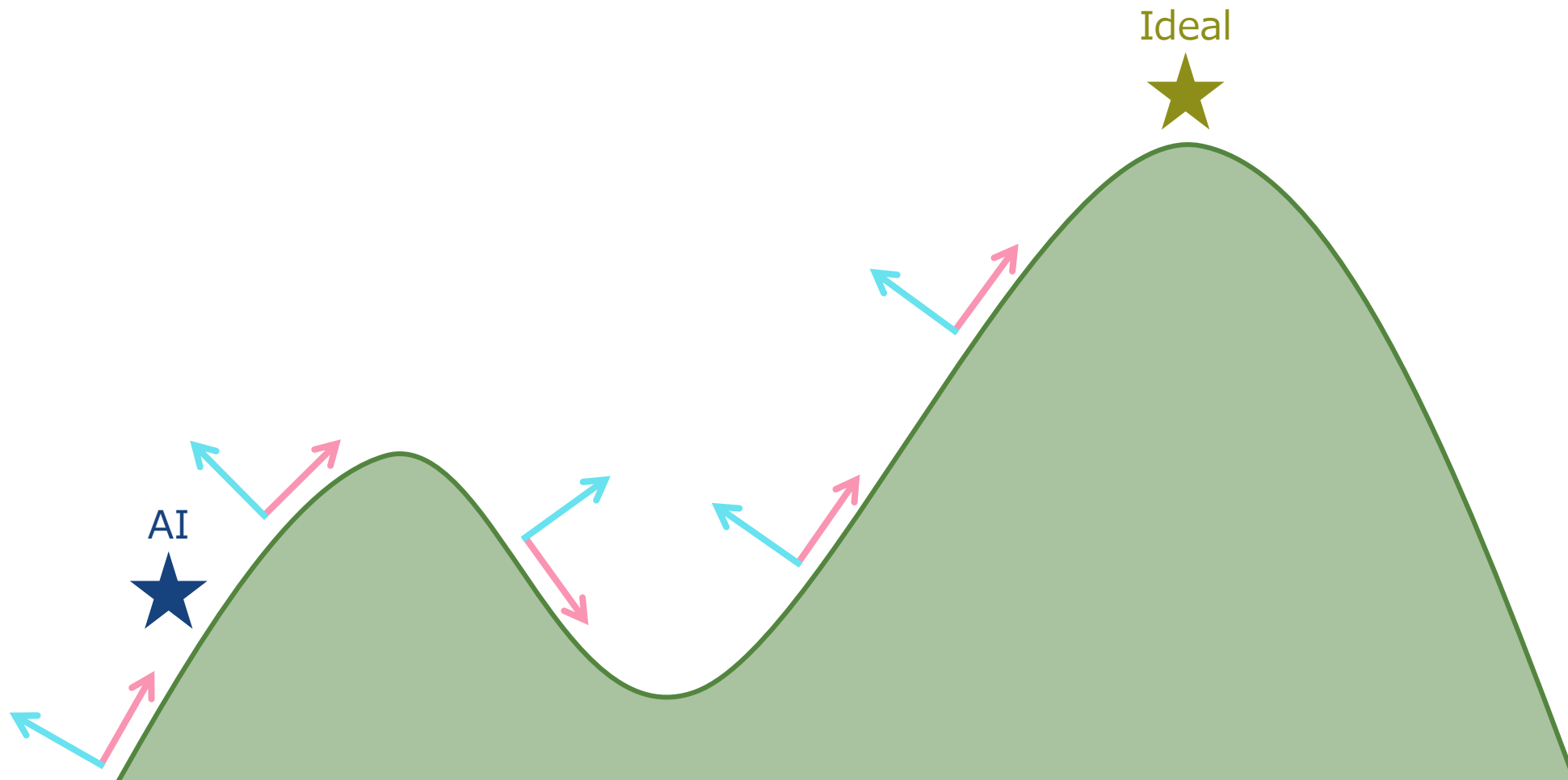
How can we design autonomous systems with guarantees?

Distance increases downhill ...



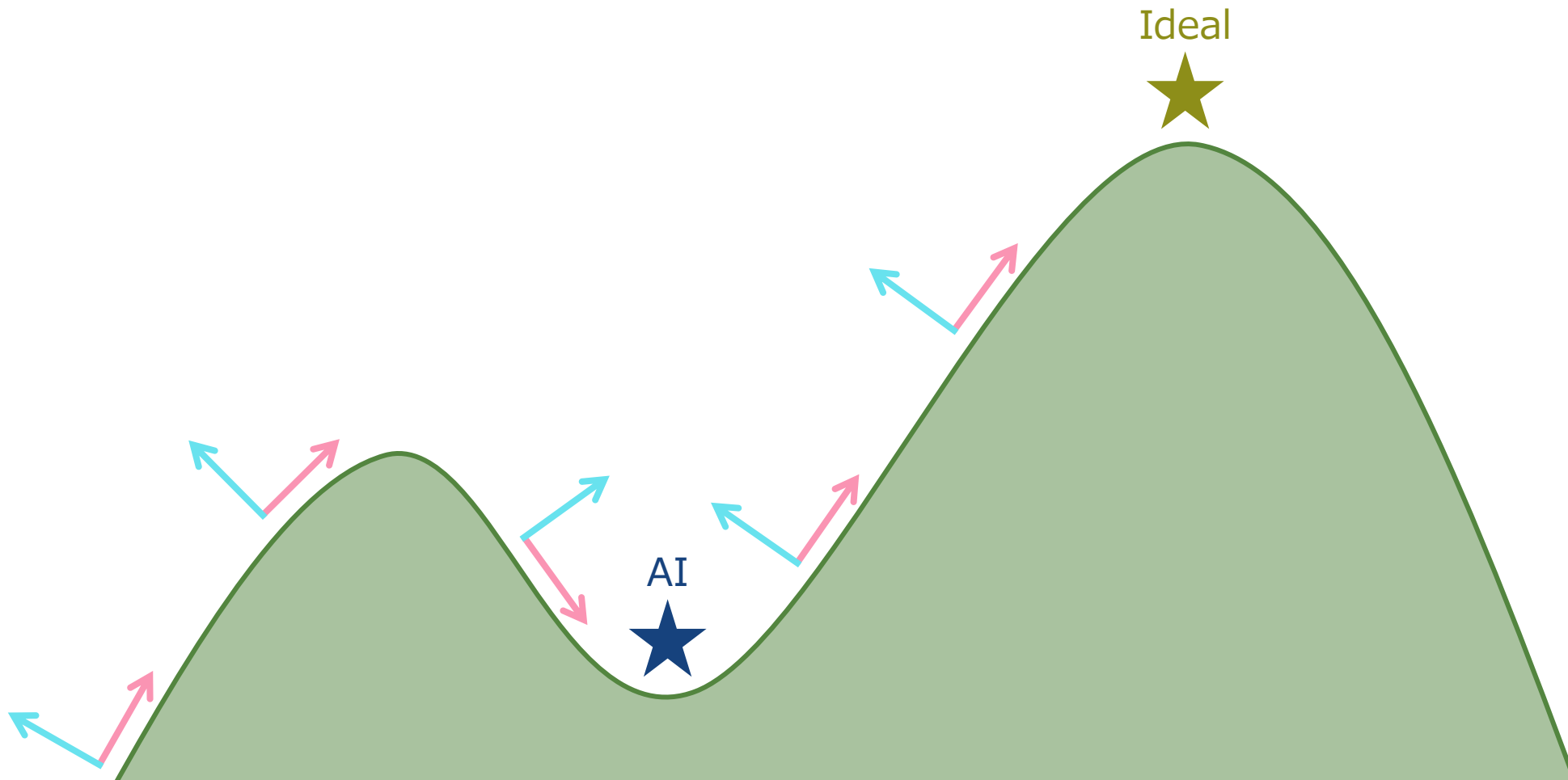
How can we design autonomous systems with guarantees?

If distance is measured along mountain slopes ...



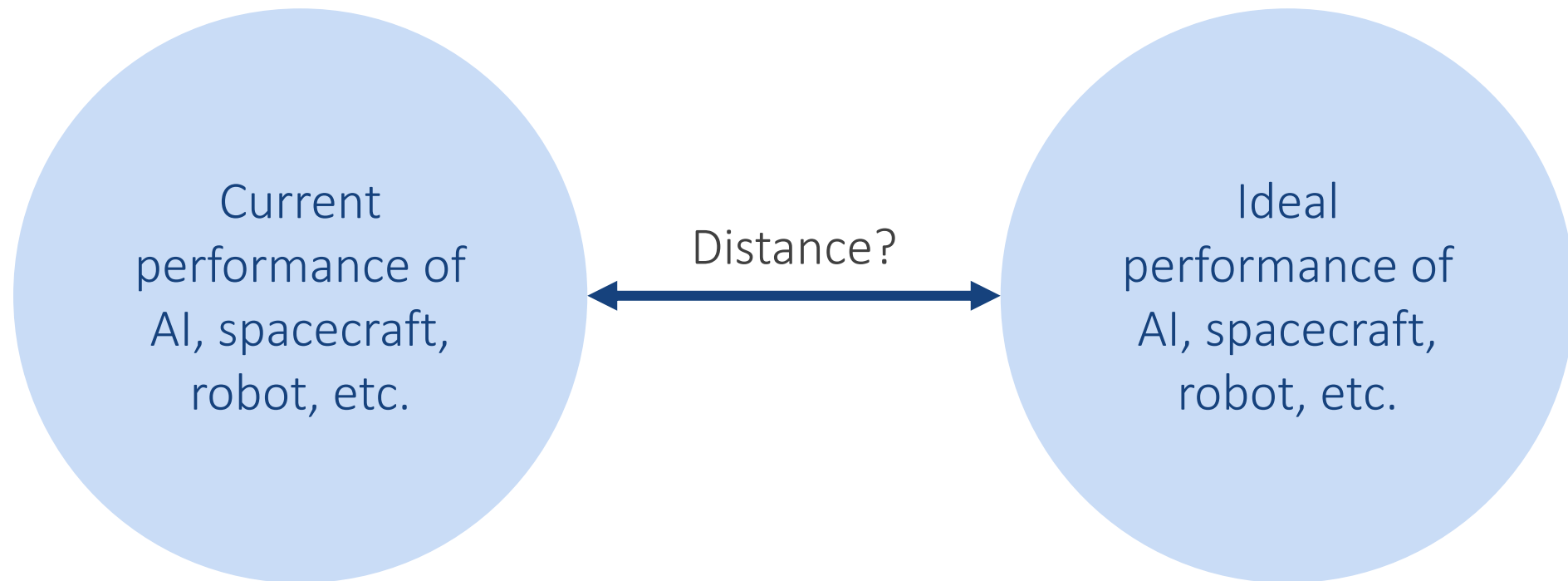
How can we design autonomous systems with guarantees?

Distance keeps decreasing even downhill!



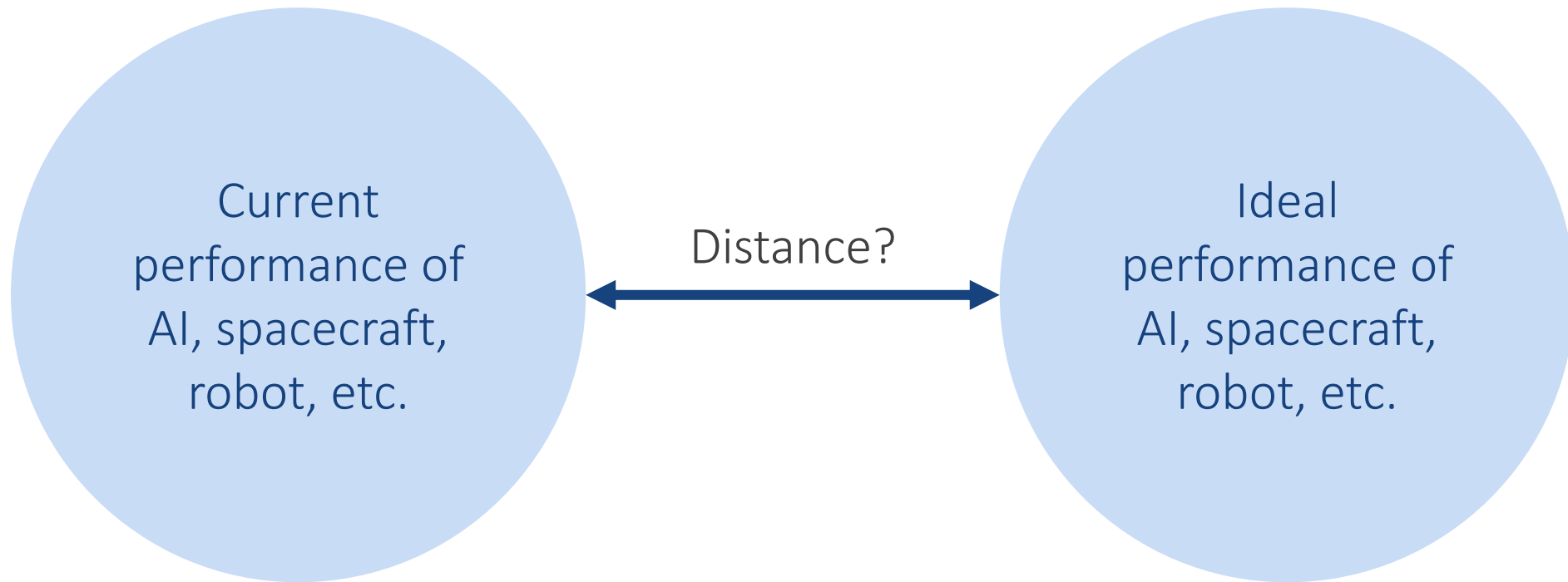
How can we design autonomous systems with guarantees?

We DEFINE a metric (i.e., how we measure a distance) so that the distance measured in the metric decreases as a result of systems' decision-making



How can we design autonomous systems with guarantees?

Contraction Theory



Robots





* This shoot was conducted with a permit. Users wishing to film in public areas should comply with the applicable regulations.

** All footage was shot by professionals in compliance with local laws and regulations. Always observe local laws and regulations and ensure the aircraft has obtained relevant certifications and airspace authorization.



Flapper Nimble Drone for Nonlinear GNC



YouTube: <https://youtu.be/YxzhluHhusM>





